

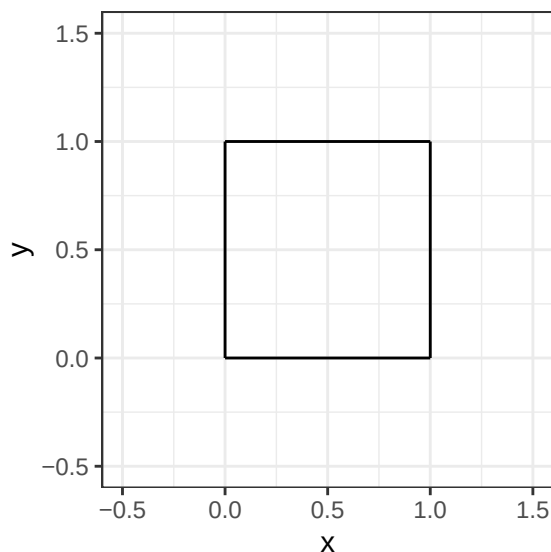
## Lab Three – Matrices and Data Frames in R

- Complete the tasks below. Make sure to start your solutions in on a new line that starts with “**Solution:**”.
- Make sure to use the Quarto Cheatsheet. This will make completing and writing up the lab *much* easier.

Consider the unit square depicted in Figure 1.

```
1 ggplot() +  
2   geom_segment(aes(x = 0, xend = 0,  
3                   y=0, yend = 1))+  
4   geom_segment(aes(x = 0, xend = 1,  
5                   y=1, yend = 1))+  
6   geom_segment(aes(x = 1, xend = 1,  
7                   y=0, yend = 1))+  
8   geom_segment(aes(x = 0, xend = 1,  
9                   y=0, yend = 0)) +  
10  theme_bw() +  
11  xlim(-0.5, 1.5)+  
12  ylim(-0.5, 1.5)+  
13  labs(x = "x",  
14       y = "y")
```

Figure 1: The unit square.



The unit square can be defined by two basis vectors

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

From these vectors, we can compute the corner vertices.

|                                |                            |
|--------------------------------|----------------------------|
| $p_0 = (0, 0) = (x_0, y_0)$    | (The origin)               |
| $p_1 = v_1 = (x_1, y_1)$       | (The first basis vector)   |
| $p_2 = v_1 + v_2 = (x_2, y_2)$ | (The sum of basis vectors) |
| $p_3 = v_1 = (x_3, y_3)$       | (The second basis vector)  |

Note that the segments that form the boundary go from  $p_0$  to  $p_1$  to  $p_2$  to  $p_3$ .

Consider the matrix  $M$  to be the matrix we get from binding the two basis vectors at the column,

$$\mathbf{M} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

## 1 Question 1

Let's transform  $\mathbf{M}$  to demonstrate an interesting linear algebra property.

### 1.1 Part a

Create the matrix  $\mathbf{A}$  in R.

$$\mathbf{A} = \begin{bmatrix} 1 & -7 \\ 5 & 9 \end{bmatrix}$$

#### Solution

```
1 (A = matrix(data=c(1,-7,5,9), nrow=2, ncol=2, byrow=TRUE))
```

```
      [,1] [,2]  
[1,]     1  -7  
[2,]     5   9
```

### 1.2 Part b

Compute the product below in R and store it in an object  $\mathbf{T}$ .

$$\mathbf{T} = \mathbf{A}\mathbf{M}.$$

#### Solution

```
1 (M = diag(2))
```

```
      [,1] [,2]  
[1,]     1   0  
[2,]     0   1
```

```
1 (T = A %*% M)
```

```
      [,1] [,2]  
[1,]     1  -7  
[2,]     5   9
```

### 1.3 Part c

Create `basis.vector.1` (first column of  $\mathbf{T}$ ) and `basis.vector.2` (second column of  $\mathbf{T}$ ) in R.

#### Solution

```
1 (basis.vector.1 = T[,1])
```

```
[1] 1 5
```

```
1 (basis.vector.2 = T[,2])
```

```
[1] -7 9
```

### 1.4 Part d

Compute the points  $p_0$ ,  $p_1$ ,  $p_2$ , and  $p_3$ .

#### Solution

```
1 (p.0 = c(0,0))
```

```
[1] 0 0
```

```
1 (p.1 = basis.vector.1)
```

```
[1] 1 5
```

```

1 (p.2 = basis.vector.1 + basis.vector.2)

[1] -6 14
1 (p.3 = basis.vector.2)

[1] -7 9

```

## 1.5 Part e

Copy and paste the code for the unit square. Edit the code to draw the segments that form the boundary based on the points in Part d.

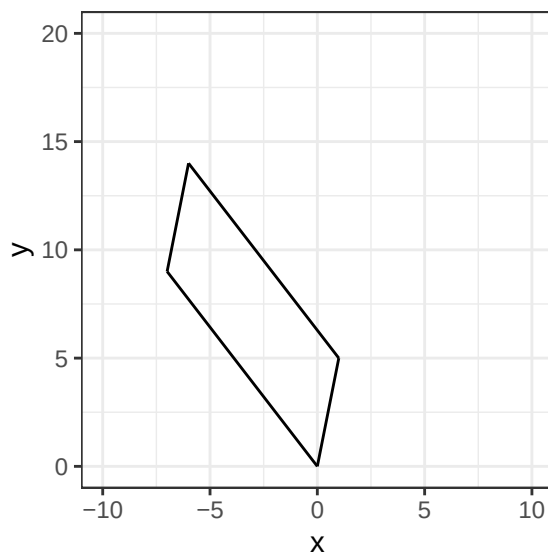
### Solution

```

1 ggplot() +
2   geom_segment(aes(x = p.0[1], xend = p.1[1],
3                     y=p.0[2], yend = p.1[2]))+
4   geom_segment(aes(x = p.1[1], xend = p.2[1],
5                     y=p.1[2], yend = p.2[2]))+
6   geom_segment(aes(x = p.2[1], xend = p.3[1],
7                     y=p.2[2], yend = p.3[2]))+
8   geom_segment(aes(x = p.3[1], xend = p.0[1],
9                     y=p.3[2], yend = p.0[2])) +
10  theme_bw() +
11  xlim(-10, 10)+
12  ylim(0, 20)+
13  labs(x = "x",
14        y = "y")

```

Figure 2: Shape (part 1e).



## 1.6 Part f

Interestingly, the determinant of  $\mathbf{A}$  gives us the area of the shape plotted in Part e. Find the determinant of  $\mathbf{A}$ .

### Solution

```

1 (det.A = det(A))

[1] 44

```

The area of the shape we plotted is 44.

## 2 Question 2

Recomplete Question 1 using a new matrix for  $\mathbf{A}$ . This matrix is not as well behaved.

## 2.1 Part a

Create the matrix  $A$  in R.

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

### Solution

```
1 (A = matrix(data=c(1,2,2,4), nrow=2, ncol=2, byrow=TRUE))
```

```
      [,1] [,2]  
[1,]    1    2  
[2,]    2    4
```

## 2.2 Part b

Compute the product below in R and store it in an object  $T$ .

$$T = AM.$$

### Solution

```
1 (M = diag(2))
```

```
      [,1] [,2]  
[1,]    1    0  
[2,]    0    1
```

```
1 (T = A %*% M)
```

```
      [,1] [,2]  
[1,]    1    2  
[2,]    2    4
```

## 2.3 Part c

Create `basis.vector.1` (first column of  $T$ ) and `basis.vector.2` (second column of  $T$ ) in R.

### Solution

```
1 (basis.vector.1 = T[,1])
```

```
[1] 1 2
```

```
1 (basis.vector.2 = T[,2])
```

```
[1] 2 4
```

## 2.4 Part d

Compute the points  $p_0$ ,  $p_1$ ,  $p_2$ , and  $p_3$ .

### Solution

```
1 (p.0 = c(0,0))
```

```
[1] 0 0
```

```
1 (p.1 = basis.vector.1)
```

```
[1] 1 2
```

```
1 (p.2 = basis.vector.1 + basis.vector.2)
```

```
[1] 3 6
```

```
1 (p.3 = basis.vector.2)
```

```
[1] 2 4
```

## 2.5 Part e

Copy and paste the code for plotting the unit square. Edit the code to draw the segments that form the boundary based on the points in Part d.

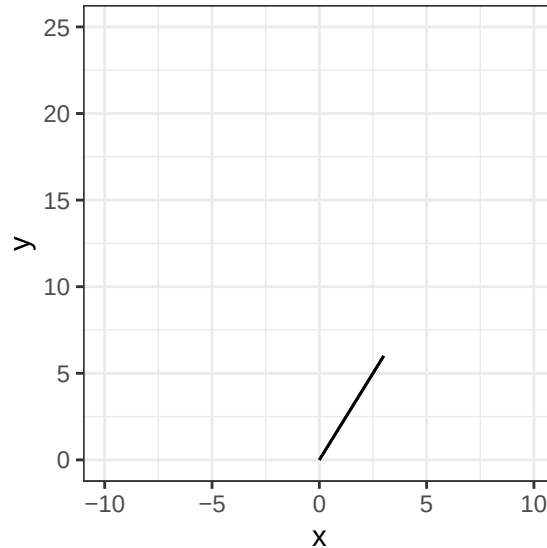
### Solution

```

1 ggplot() +
2   geom_segment(aes(x = p.0[1], xend = p.1[1],
3                     y=p.0[2], yend = p.1[2]))+
4   geom_segment(aes(x = p.1[1], xend = p.2[1],
5                     y=p.1[2], yend = p.2[2]))+
6   geom_segment(aes(x = p.2[1], xend = p.3[1],
7                     y=p.2[2], yend = p.3[2]))+
8   geom_segment(aes(x = p.3[1], xend = p.0[1],
9                     y=p.3[2], yend = p.0[2])) +
10  theme_bw() +
11  xlim(-10, 10)+
12  ylim(0, 25)+
13  labs(x = "x",
14        y = "y")

```

Figure 3: Shape (part 2e).



## 2.6 Part f

Interestingly, the determinant of  $\mathbf{A}$  gives us the area of the shape plotted in Part e. Find the determinant of  $\mathbf{A}$ .

### Solution

```

1 (det.A = det(A))

```

```
[1] 0
```

The area of the shape we plotted is 0.

## 3 Question 3

We can conduct a shear transformation that keeps the area of the shape the same but horizontally or vertically slants the object.

In fact, this is one of the ways artists can make something look like it's leaning or being pushed. Other applications where this is useful include fluid dynamics and crystallography, where the area of a deformed object must remain constant, or in photography where perspective in photos can be adjusted.

A shear transformation is implemented by altering the basis vectors we start with. For vertical slanting,

$$\mathbf{M}_v = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$$

and for horizontal slanting

$$\mathbf{M}_h = \begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}.$$

In both cases,  $k \in \mathbb{R}$  is called a shear factor and its sign determines the direction of the slant and its magnitude determines the severity of the slant.

### 3.1 Part a

Use the following to generate a new shape. Compare the shape and its area to your answer in Question 1.

$$\mathbf{M}_v = \begin{bmatrix} 1 & k = 0.5 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{A} = \begin{bmatrix} 1 & -7 \\ 5 & 9 \end{bmatrix}$$

#### Solution

```

1 k = 0.5
2 (M.v = matrix(data=c(1,k,0,1),nrow=2,ncol=2,byrow=TRUE))

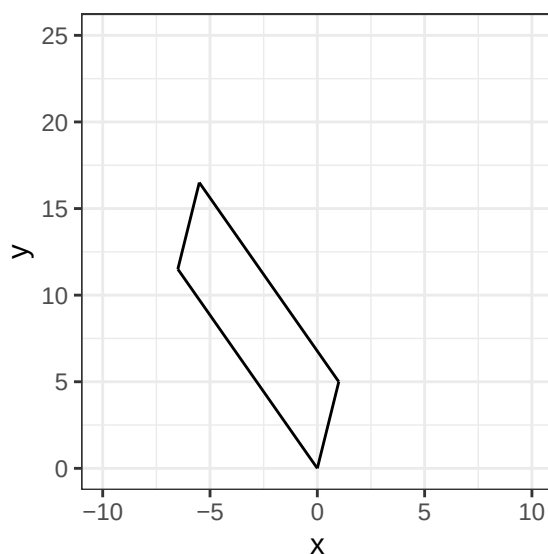
      [,1] [,2]
[1,]    1  0.5
[2,]    0  1.0
1 (A = matrix(data=c(1,-7,5,9),nrow=2,ncol=2,byrow=TRUE))

      [,1] [,2]
[1,]    1  -7
[2,]    5   9
1 T = A %*% M.v
2
3 basis.vector.1 = T[,1]
4 basis.vector.2 = T[,2]
5
6 p.0 = c(0,0)
7 p.1 = basis.vector.1
8 p.2 = basis.vector.1 + basis.vector.2
9 p.3 = basis.vector.2

1 ggplot() +
2   geom_segment(aes(x = p.0[1], xend = p.1[1],
3                     y=p.0[2], yend = p.1[2]))+
4   geom_segment(aes(x = p.1[1], xend = p.2[1],
5                     y=p.1[2], yend = p.2[2]))+
6   geom_segment(aes(x = p.2[1], xend = p.3[1],
7                     y=p.2[2], yend = p.3[2]))+
8   geom_segment(aes(x = p.3[1], xend = p.0[1],
9                     y=p.3[2], yend = p.0[2])) +
10  theme_bw() +
11  xlim(-10, 10)+
12  ylim(0, 25)+
13  labs(x = "x",
14        y = "y")

```

Figure 4: Shape (part 3a).



```
1 (det.A = det(A))
```

```
[1] 44
```

The shape we generate is a parallelogram, very similar to what we generated in part 1e; however, it is vertically slanted. This is obviously very different from the line segment produced in 2e.

The area of the shape generated for 3a is the same as the area of the shape we generated in 1e. This is because the shear transformation preserves area.

## 3.2 Part b

Use the following to generate a new shape. Compare the shape and its area to your answer in Questions 1 and the vertical shear transform in part (a).

$$\mathbf{M}_h = \begin{bmatrix} 1 & 0 \\ k = 0.5 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{A} = \begin{bmatrix} 1 & -7 \\ 5 & 9 \end{bmatrix}$$

### Solution

```
1 k = 0.5
2 (M.v = matrix(data=c(1,0,k,1),nrow=2,ncol=2,byrow=TRUE))
```

```
      [,1] [,2]
[1,]  1.0   0
[2,]  0.5   1
```

```
1 (A = matrix(data=c(1,-7,5,9),nrow=2,ncol=2,byrow=TRUE))
```

```
      [,1] [,2]
[1,]    1  -7
[2,]    5   9
```

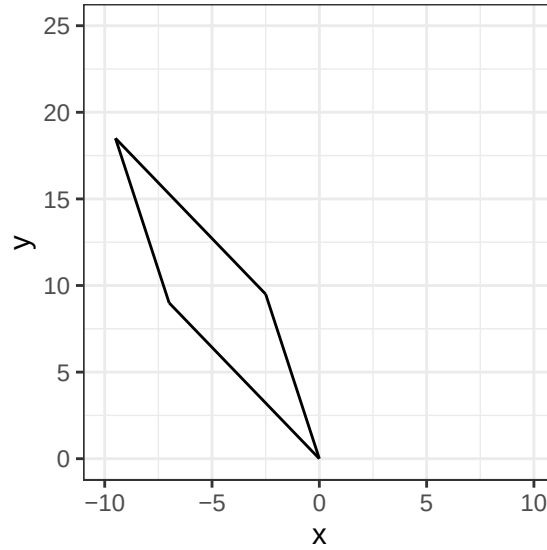
```
1 T = A %*% M.v
```

```
2
3 basis.vector.1 = T[,1]
4 basis.vector.2 = T[,2]
```

```
5
6 p.0 = c(0,0)
7 p.1 = basis.vector.1
8 p.2 = basis.vector.1 + basis.vector.2
9 p.3 = basis.vector.2
```

```
1 ggplot() +
2   geom_segment(aes(x = p.0[1], xend = p.1[1],
3                     y=p.0[2], yend = p.1[2]))+
4   geom_segment(aes(x = p.1[1], xend = p.2[1],
5                     y=p.1[2], yend = p.2[2]))+
6   geom_segment(aes(x = p.2[1], xend = p.3[1],
7                     y=p.2[2], yend = p.3[2]))+
8   geom_segment(aes(x = p.3[1], xend = p.0[1],
9                     y=p.3[2], yend = p.0[2])) +
10  theme_bw() +
11  xlim(-10, 10)+
12  ylim(0, 25)+
13  labs(x = "x",
14       y = "y")
```

Figure 5: Shape (part 3b).



```
1 (det.A = det(A))
```

```
[1] 44
```

The shape generated for *3b* resembles the shape from *1e* but slanted horizontally, retaining the same area. This is what one would expect based on the shear transformation. It begins to take on more of a diamond/kite shape when compared to *1e* and *3a*, though it is still a parallelogram. Horizontal slants of a parallelogram seem to be more pronounced than vertical slants of the same magnitude.

### 3.3 Part c

Can we shear vertically and horizontally at the same time? Use the following to generate a new shape. Compare the shape and its area to your answers in parts (a) and (b).

$$\mathbf{M}_{vh} = \begin{bmatrix} 1 & k = 0.5 \\ k = 0.5 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{A} = \begin{bmatrix} 1 & -7 \\ 5 & 9 \end{bmatrix}$$

#### Solution

```
1 k = 0.5
2 (M.v = matrix(data=c(1,k,k,1),nrow=2,ncol=2,byrow=TRUE))
```

```
      [,1] [,2]
[1,]  1.0  0.5
[2,]  0.5  1.0
```

```
1 (A = matrix(data=c(1,-7,5,9),nrow=2,ncol=2,byrow=TRUE))
```

```
      [,1] [,2]
[1,]    1  -7
[2,]    5   9
```

```
1 T = A %*% M.v
```

```
2
3 basis.vector.1 = T[,1]
4 basis.vector.2 = T[,2]
```

```
5
6 p.0 = c(0,0)
7 p.1 = basis.vector.1
8 p.2 = basis.vector.1 + basis.vector.2
9 p.3 = basis.vector.2
```

```
1 ggplot() +
2   geom_segment(aes(x = p.0[1], xend = p.1[1],
3                     y = p.0[2], yend = p.1[2]))+
4   geom_segment(aes(x = p.1[1], xend = p.2[1],
5                     y = p.1[2], yend = p.2[2]))+
6   geom_segment(aes(x = p.2[1], xend = p.3[1],
7                     y = p.2[2], yend = p.3[2]))+
```

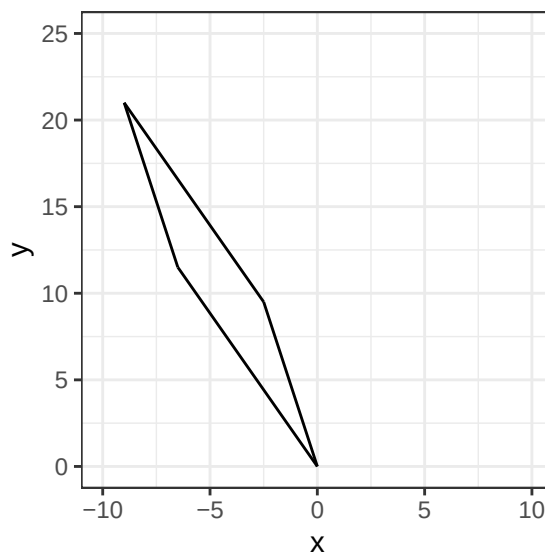


```

8   geom_segment(aes(x = p.3[1], xend = p.0[1],
9                     y=p.3[2], yend = p.0[2])) +
10  theme_bw() +
11  xlim(-10, 10)+
12  ylim(0, 25)+
13  labs(x = "x",
14        y = "y")

```

Figure 6: Shape (part 3c).



```

1   (det.A = det(A))

```

[1] 44

The area, again, is unsurprisingly the same as  $1e$  and its subsequent transformations ( $3a$  and  $3b$ ). This transformation takes on characteristics of both  $3a$  and  $3b$ , being the skinniest and longest of the generations.

## 4 Question 4

### 4.1 Part a

Create a data frame `ladder` with the following data. Note your data frame should have three columns; I split it to save vertical space here.

| x | y  | group | x | y  | group |
|---|----|-------|---|----|-------|
| 0 | 0  | 1     | 1 | 8  | 6     |
| 0 | 20 | 1     | 0 | 10 | 7     |
| 1 | 0  | 2     | 1 | 10 | 7     |
| 1 | 20 | 2     | 0 | 12 | 8     |
| 0 | 2  | 3     | 1 | 12 | 8     |
| 1 | 2  | 3     | 0 | 14 | 9     |
| 0 | 4  | 4     | 1 | 14 | 9     |
| 1 | 4  | 4     | 0 | 16 | 10    |
| 0 | 6  | 5     | 1 | 16 | 10    |
| 1 | 6  | 5     | 0 | 18 | 11    |
| 0 | 8  | 6     | 1 | 18 | 11    |

**Solution**

```

1 (ladder = data.frame(x=c(0,0,1,1,0,1,0,1,0,1,0,1,0,1,0,1,0,1,0,1,0,1),y=c(0,20,0,20,2,2,4,4,6,6,8,8,10,10,12,12,14,14,16,16,18,18),group=c(1,1,2,2,3,3,
  x y group
1 0 0 1
2 0 20 1
3 1 0 2
4 1 20 2
5 0 2 3
6 1 2 3
7 0 4 4
8 1 4 4
9 0 6 5
10 1 6 5
11 0 8 6
12 1 8 6
13 0 10 7
14 1 10 7
15 0 12 8
16 1 12 8
17 0 14 9
18 1 14 9
19 0 16 10
20 1 16 10
21 0 18 11
22 1 18 11

```

## 4.2 Part b

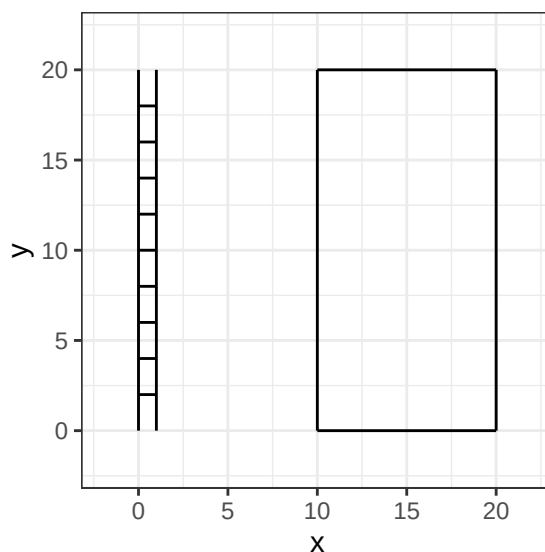
Remove `#|eval: false` in the code below. The result should look like a ladder and a very rectangular house if part a is correct.

```

1 ggplot() +
2   geom_path(data = ladder,
3             aes(x = x, y = y, group = group)) +
4     geom_segment(aes(x = 10, xend = 10,
5                     y=0, yend = 20))+
6   geom_segment(aes(x = 10, xend = 20,
7                   y=20, yend = 20))+
8   geom_segment(aes(x = 20, xend = 20,
9                   y=0, yend = 20))+
10  geom_segment(aes(x = 10, xend = 20,
11                  y=0, yend = 0)) +
12  theme_bw() +
13  xlim(-2,22) +
14  ylim(-2,22)

```

Figure 7: A ladder.



### 4.3 Part c

Use `as.matrix()` to create a  $22 \times 2$  matrix **A** containing the `x` and `y` observations from the `ladder` data frame. Use matrix multiplication to compute

$$\mathbf{T} = \mathbf{A}\mathbf{M}_h.$$

Recall,

$$\mathbf{M}_h = \begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$$

and use  $k = 0.5$ .

#### Solution

```
1 k=0.5
2 (M.h = matrix(data=c(1,0,k,1),nrow=2,ncol=2,byrow=TRUE))
```

```
      [,1] [,2]
[1,]  1.0   0
[2,]  0.5   1
```

```
1 (A = as.matrix(ladder[,c(1,2)]))
```

```
      x  y
[1,] 0  0
[2,] 0 20
[3,] 1  0
[4,] 1 20
[5,] 0  2
[6,] 1  2
[7,] 0  4
[8,] 1  4
[9,] 0  6
[10,] 1  6
[11,] 0  8
[12,] 1  8
[13,] 0 10
[14,] 1 10
[15,] 0 12
[16,] 1 12
[17,] 0 14
[18,] 1 14
[19,] 0 16
[20,] 1 16
[21,] 0 18
[22,] 1 18
```

```
1 (T = A %*% M.h)
```

```
      [,1] [,2]
[1,]    0    0
[2,]   10   20
[3,]    1    0
[4,]   11   20
[5,]    1    2
[6,]    2    2
[7,]    2    4
[8,]    3    4
[9,]    3    6
[10,]   4    6
[11,]   4    8
[12,]   5    8
[13,]   5   10
[14,]   6   10
[15,]   6   12
[16,]   7   12
[17,]   7   14
[18,]   8   14
[19,]   8   16
[20,]   9   16
[21,]   9   18
[22,]  10   18
```

### 4.4 Part d

Create a data frame `leaning.ladder` with the `x` and `y` equal to the first and second column of **T**, respectively, and the same values of `group` from `ladder`.

#### Solution

```
1 (leaning.ladder = data.frame(x = T[,1],y=T[,2],group=ladder[,3]))
```

|    | x  | y  | group |
|----|----|----|-------|
| 1  | 0  | 0  | 1     |
| 2  | 10 | 20 | 1     |
| 3  | 1  | 0  | 2     |
| 4  | 11 | 20 | 2     |
| 5  | 1  | 2  | 3     |
| 6  | 2  | 2  | 3     |
| 7  | 2  | 4  | 4     |
| 8  | 3  | 4  | 4     |
| 9  | 3  | 6  | 5     |
| 10 | 4  | 6  | 5     |
| 11 | 4  | 8  | 6     |
| 12 | 5  | 8  | 6     |
| 13 | 5  | 10 | 7     |
| 14 | 6  | 10 | 7     |
| 15 | 6  | 12 | 8     |
| 16 | 7  | 12 | 8     |
| 17 | 7  | 14 | 9     |
| 18 | 8  | 14 | 9     |
| 19 | 8  | 16 | 10    |
| 20 | 9  | 16 | 10    |
| 21 | 9  | 18 | 11    |
| 22 | 10 | 18 | 11    |

## 4.5 Part e

Remove `eval: false` in the code below. The result should look like a ladder and a very rectangular house if part a is correct. The new leaning ladder should look like it is leaning on the house.

```

1 ggplot() +
2   geom_path(data = ladder,
3             aes(x = x, y = y, group = group, color = "Starting Ladder")) +
4   geom_path(data = leaning.ladder,
5             aes(x = x, y = y, group = group, color = "Leaning Ladder")) +
6   geom_segment(aes(x = 10, xend = 10,
7                    y=0, yend = 20))+
8   geom_segment(aes(x = 10, xend = 20,
9                    y=20, yend = 20))+
10  geom_segment(aes(x = 20, xend = 20,
11                  y=0, yend = 20))+
12  geom_segment(aes(x = 10, xend = 20,
13                  y=0, yend = 0)) +
14  theme_bw() +
15  xlim(-2,22) +
16  ylim(-2,22) +
17  scale_color_manual("", values=c("darkred", "grey"))

```

Figure 8: A ladder and a leaning ladder.

